ANSWER KEY & MARKING SCHEME SET A

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** palindrome	e.g. 121
2	**(a)** an unsolved problem	mystery
3	**(c)** 8531	descending digits
4	**(b)** 3500	estimation
5	**(a)** 5138	last digit 8
6	**(b)** calculating quickly in the mind using tricks	easy strategies
7	**(b)** 99999	all 9s
8	**(a)** odd	e.g. 4+5=9
9	**(d)** 6	apply the rule repeatedly
10	**(b)** 3087	largest minus smallest

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A number or word reading the same both ways; 121 and 4554.	1+1
12	9642 and 2469.	1+1
13	5431 − 1345 = 4086.	1+1
14	5, 16, 8, 4, 2, 1.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	8730 − 0378 = 8352 (1); 8532 − 2358 = 6174 (1); then it stays at 6174 (1).	1+1+1
16	7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (2); 16 steps (1).	2+1
17	e.g. 1221, 3443; each palindrome is its own reverse (1.5); sum is twice the number (1.5).	1½+1½
	OR: 500 × 50 = 25,000 (1.5); exact 25,324 (1.5).	1½+1½
18	Take 68 + 86 = 154; 154 + 451 = 605; 605 + 506 = 1111 (2); a palindrome (1).	2+1
	OR: Sum is 45 = 3 × 15 (2); it equals 3 times the middle number (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Rule (1). (ii) 5321 − 1235 = 4086 (2). (iii) 4086 → 8640 − 0468 = 8172 → 8721 − 1278 = 7443 → 7443 − 3447 = 3996 → 9963 − 3699 = 6264 → 6642 − 2466 = 4176 → 7641 − 1467 = 6174 (2).	1+2+2
	OR: (i) Rules (2). (ii) 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (2). (iii) 14 steps (1).	2+2+1
20	(i) Numbers that read the same both ways (1). (ii) 161, 242, 323, 404 (2). (iii) 247 + 742 = 989 (2).	1+2+2
	OR: (i) 60 minutes (1). (ii) 10:00 (1). (iii) 4 times counting 9:00 (1). (iv) Common multiples (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 6174	reached by the Kaprekar routine
2	**(b)** 8	count of moves
3	**(c)** 22	4+5+6+7
4	**(a)** 7000	estimation
5	**(a)** from left to right	order
6	**(a)** 1089	famous trick
7	**(d)** 4	four-digit
8	**(b)** divisible by 5	last digit rule
9	**(b)** 9	apply the rule repeatedly
10	**(b)** 4086	largest minus smallest

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	5000 + 2000 = 7000.	1+1
12	It says that every starting number eventually reaches 1 with the rule; it is unproved.	1+1
13	Even; e.g. 3 + 5 = 8.	1+1
14	7,500 and 7,000.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	532 − 235 = 297 (1); 297 + 792 = 1089 (2).	1+2
16	18 (Feb) + 31 + 30 + 10 = 89 days (3).	1+2
17	Take 1 stick first, leaving 20, a multiple of 4 (1.5); after each move of the opponent take 4 minus that number (1.5).	1½+1½
	OR: 548,000 (1.5); 550,000 (1.5).	1½+1½
18	97310 and 10379 (2); 86,931 (1).	2+1
	OR: Rows, columns and both diagonals must have equal sums (1.5); the centre number is 5 (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 4000 + 5000 + 1000 = 10,000 (1.5). (ii) 50 × 20 = 1000 (1.5). (iii) 10,007 and 1,029 (2).	1½+1½+2
	OR: (i) Steps (2). (ii) 721 − 127 = 594; 594 + 495 = 1089 (2). (iii) The difference is 0 (1).	2+2+1
20	(i) 8+16 = 24, 9+15 = 24 (1). (ii) Equal (1). (iii) 12, 13, 19, 20: 32 and 32 (1). (iv) Moving down adds 7 and right adds 1 (2).	1+1+1+2
	OR: (i) Break up, round and adjust (2). (ii) 175 (1.5). (iii) 240 (1.5).	2+1½+1½

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.